

bandwidth radio channels that can support data access speeds of up to 10Gbps. The connection essentially comprises short wireless links on the end of the local fibre-optic cable. It would be more a nomadic service (like Wi-Fi) rather than a wide area mobile service.

5G mobile systems overview

The 5G technology for mobile systems is very much in the early development stages. Many companies are looking into technologies that could be used to become part of the system.

In addition to this, a number of universities have set up 5G research units focused on developing technologies for 5G. Many technologies to be used for 5G will start to appear in the systems used for 4G and then, as the new 5G mobile system starts to formulate in a more concrete manner, these will be incorporated into the new 5G mobile system.

As different generations of mobile telecommunications have evolved, each one has brought its own improvements. The same will be true of 5G technology. As with any new generation, 5G mobile technology would need to provide significant gains over previous systems to provide an adequate business case for mobile operators to invest in it.

Facilities that might be seen with 5G technology include far better levels of connectivity and coverage. The term World Wide Wireless Web, or WWW, is being coined for this. For 5G technology to be able to achieve this, new methods of connecting will be required, as one of the main drawbacks with previous generations is lack of coverage, dropped calls and low performance at cell edges. 5G technology will need to address this.

Although the standards bodies have not yet defined the parameters needed to meet a 5G performance

SOME SUGGESTED TYPICAL WIRELESS PARAMETERS FOR 5G STANDARD

Parameter	Suggested performance
Network capacity	10,000 times of the current network
Peak data rate	10Gbps
Cell edge data rate	100Mbps
Latency	<1ms

level yet, other organisations have set their own aims that may eventually influence final specifications.

These are some ideas being put forward for a 5G standard, but these are not accepted by any official body yet. The table below shows some suggested typical wireless parameters for a 5G standard.

There are many new concepts that are being investigated and developed for the new 5G mobile system. Some of these include:

Pervasive networks. This technology is being considered for 5G mobile systems, in which a user can concurrently be connected to several wireless access technologies and seamlessly move between these.

Group cooperative relay. This is a technique that is being considered to make high data rates available over a wider area of the cell. Currently, data rates fall towards the cell edge where interference levels are higher and signal levels are lower.

Cognitive radio technology. If cognitive radio technology was used for 5G mobile systems, it would enable the user equipment/handset to look at the radio landscape in which it is located and choose the optimum radio access network, modulation scheme and other parameters to configure itself to gain the best connection and optimum performance.

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